

Documentation

Antidoc v3.0.0, Diego Torres

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Chapter 1. Project description

No description found (add content in project description)

Chapter 2. DQMH® modules

This section describes DQMH® module responsibilities and relationships.

2.1. Preamble

A DQMH module is the main component of an architecture based on DQMH® framework. A DQMH module is used to implement a section of the application that has one responsibility.

DQMH® framework defines two different type of DQMH module.

Singleton:

A Singleton DQMH module can have only one instance running at any given time.

Cloneable:

A Cloneable DQMH module can have one or multiple instances running in parallel.

DQMH® framework defines two different ways to carry data throughout the application and with both other DQMH modules and non-DQMH based code.

Request events:

A request is a code that fires an event requesting the DQMH module to do something. Multiple locations in the code can send events to the DQMH module.

Request events are many-to-one.

Requests are usually named using imperative tense.

Broadcast events:

A broadcast is a code that fires an event broadcasting that the DQMH module did something. Multiple Event Structures can register to handle the Broadcast Events.

Broadcast Events are one-to-many.

Broadcasts are usually named using past tense or passive voice.

Default events:

By default all the DQMH ® Modules are shipped with some default events (Requests and Broadcasts). Here is the list of this events:

- Start Module
- Stop Module
- Show Panel
- Hide Panel
- Show Diagram
- Status Updated
- Error Reported
- Module Did Stop
- Update Module Execution Status
- Module Did Init
- Get Module Execution Status

If you want to learn more about default DQMH ® Events see the [documentaiton](#).

NOTE

Refer to the DQMH® framework official [documentation](#) to find more details on how the framework works

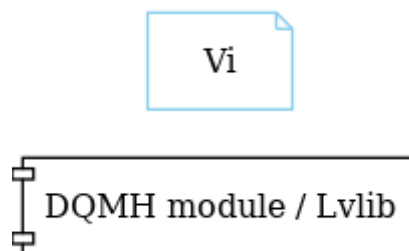
The following section gives you details on the project architecture relying on this framework. It gives you an overview of the modules' interaction and detailed information on each module.

NOTE

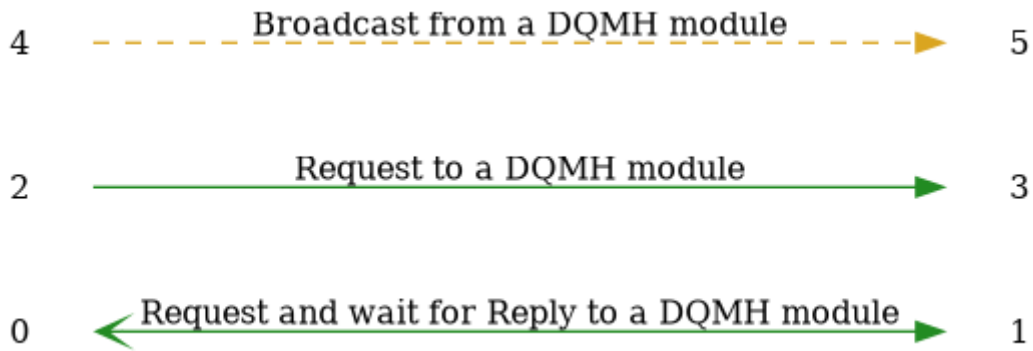
If your module has a helper loop, it will be listed along with the events it processes. Helper loops can be categorized as either DQMH or Custom. To ensure your helper loop adheres to DQMH guidelines see the [documentaiton](#).

Graphs used in this section have the following legend:

Components:



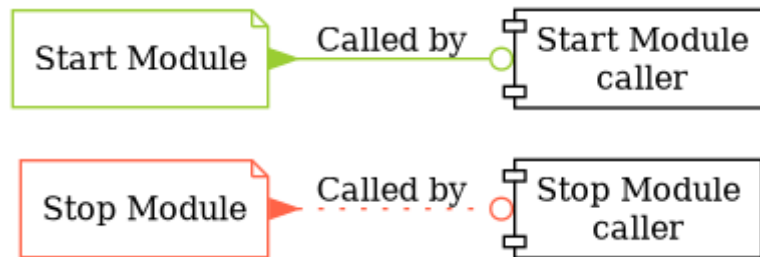
Events:



NOTE One arrow can represent one or more events between two components

NOTE Request and Request and wait for Reply are represented by only one arrow. If there is no Request and wait for Reply, Request representation is used. Otherwise Request and wait for Reply is used

Start and Stop module callers:



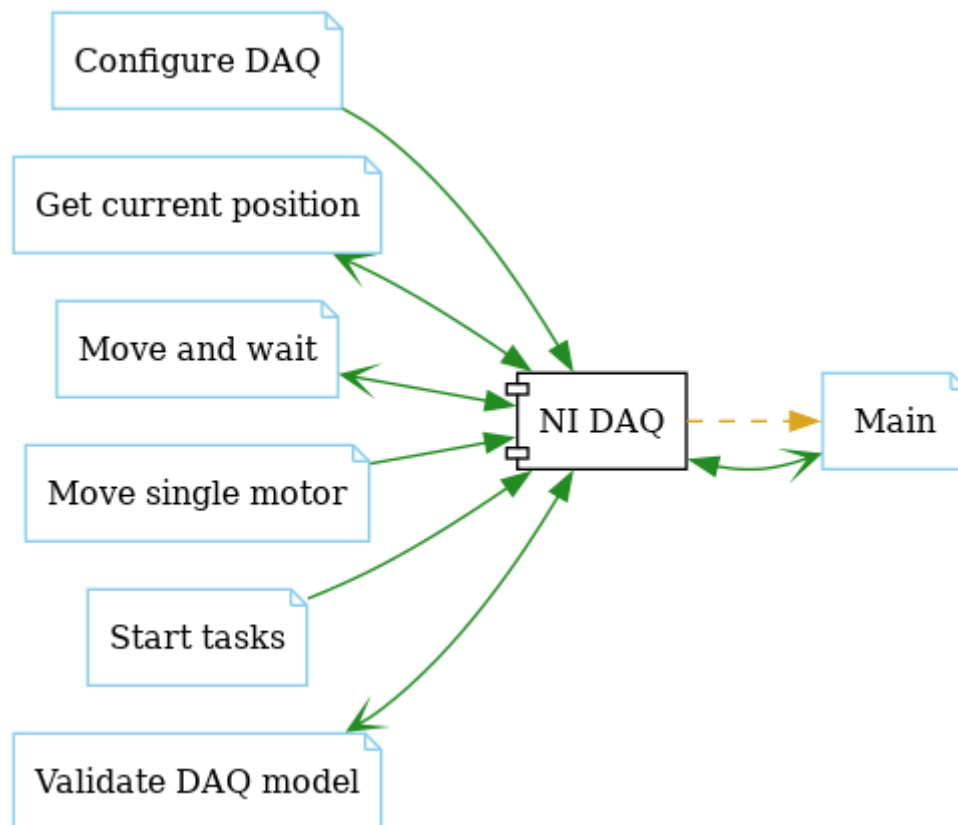
2.2. Modules overview

This project contains 0 singleton module and 1 cloneable module.

Table 1. Modules list

Singleton	Cloneable
	NI DAQ.lvlib

This graph represents the links between all DQMH modules.



2.3. NI DAQ.lvlib

Type: Cloneable

Responsibility: Management of connection, aquisition and signal generation from NI DAQ devices.

2.3.1. Module relationship

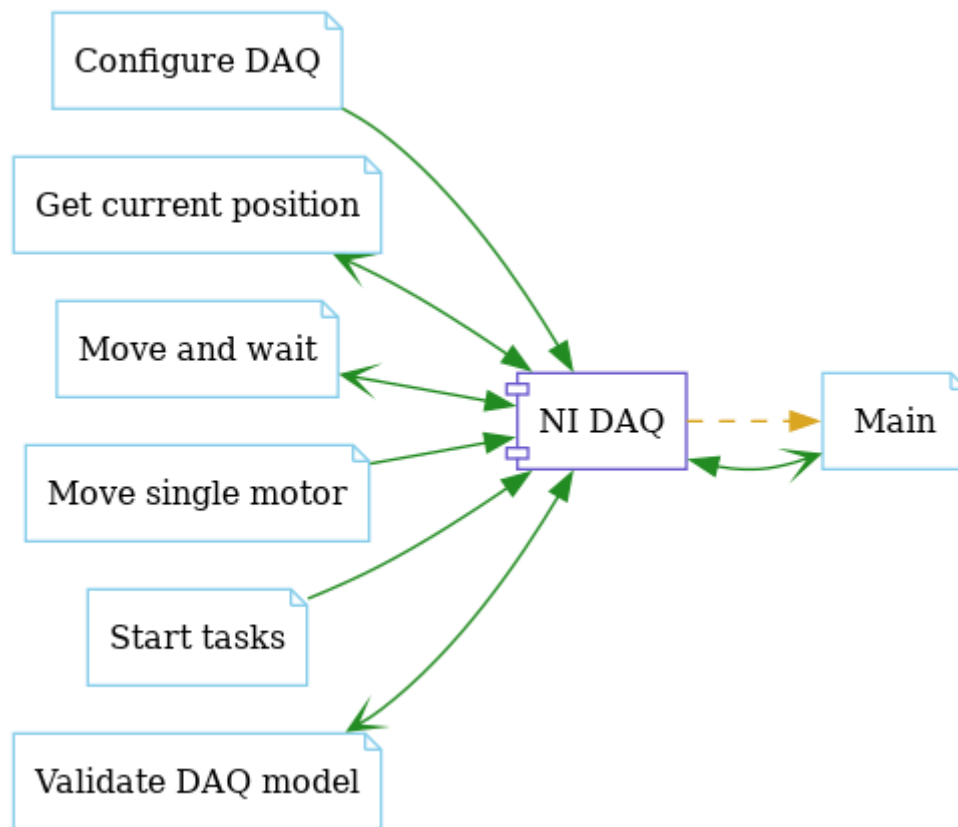


Table 2. Requests callers

Request Name	Callers
Create AI channel	Configure DAQ.vi
Create AO channel	Configure DAQ.vi
Create CI channel	Configure DAQ.vi
Create DI channel	Configure DAQ.vi
Create DO channel	Configure DAQ.vi
Get channel list	
Get device model	Validate DAQ model.vi
Get device name	Main.vi
Get task ref	
Hide Panel	
Read AI	Main.vi
Read CI	Get current position.vi Main.vi Move and wait.vi
Read DI	
Set device name	Main.vi Validate DAQ model.vi
Show Diagram	

Request Name	Callers
Show Panel	
Sleep HL	
Start continous acquisition	Start tasks.vi
Start task	Main.vi Move and wait.vi Start tasks.vi
Stop continuous acquisition	Main.vi
Stop task	Main.vi Move and wait.vi
Validate connection	Main.vi
Wake up HL	
Write AO	Main.vi Start tasks.vi
Write DO	Main.vi Move and wait.vi Move single motor.vi Start tasks.vi

Table 3. Broadcasts Listeners

Broadcast Name	Listeners
Analog data acquired	
Connection failed	Main.vi
Counter data acquired	
Device connected	Main.vi
Digital data acquired	Main.vi
Error Reported	
Module Did Init	Main.vi
Module Did Stop	
Status Updated	
Update Module Execution Status	

Table 4. Used requests

Module	Requests
—	—

Table 5. Registered broadcast

Module	Broadcasts
—	—

2.3.2. Event list

NOTE The [default events](#) are not listed.

Table 6. Events

Name	Type	Connector pane	Description	S.	R.	I.
Set device name	→		Set the device identifier to be managed internally by the module.			
Get channel list	→		Request to read all available channels from the specified device.			
Start task	→		Request to start all existing channels for the local task.			
Stop task	→		Request to stop all channels within the local task.			
Write DO	→		Request to write to a single or multiple digital output channels.			
Create DI channel	→		Request to create a digital input channel(s).			
Read DI	→		Request to read one or multiple digital inputs.			
Get task ref	→		Request to get a reference to a given task.			
Wake up HL	→		Request to wake the helper loop to continuously read/acquire from the specified task at a rate determined by the timeout. Returns data as a variant.			
Sleep HL	→		Request to stop the helper loop and continuous acquisition.			
Create AI channel	→		Request to create an analog input channel in the selected channel.			
Read AI	→		Request to read single or multiple analog inputs.			
Create AO channel	→		Request to create single or multiple analog output channel (s).			

Name	Type	Connector pane	Description	S.	R.	I.
Write AO	→		Request to write a value to a single or multiple channel (s)			
Validate connection	→		Request to validate correct connection to device. Call after the Set device ID event.			
Start continuous acquisition	→		Request to acquire continuously from DAQ device with a predefined Timeout period.			
Stop continuous acquisition	→		Request to stop continuous acquisition made by the HL.			
Create DO channel	→		Creates a new digital output with a single or multiple channels.			
Create CI channel	→		Request to create a counter input channel.			
Read CI	→		Request to read the number of pulses of CI channels.			
Get device model	→		Request to get the model of the device.			
Get device name	→		Request to return the name of the device name set to the module.			
Digital data acquired	→		Note: This VI was renamed by the DQMH Rename Event utility. Make sure the VI Description is updated to reflect the new event name, then delete this comment. Broadcasts data acquired from a digital or analog channel.			
Device connected	→		Broadcasts whenever connection to a DAQ device has been achieved successfully.			
Connection failed	→		Broadcasts whenever an unsuccessful attempt to connect has been made.			
Analog data acquired	→		Broadcasts data from an AI task.			
Counter data acquired	→		Reports whenever the input counters have information available.			

Type: ➡ → Request | ➡➡ → Request and Wait for Reply | ➡➡➡ → Broadcast

Scope: 🔑 → Protected | 🔑➡ → Community

Reentrancy: 📄 → Preallocated reentrancy | 📄📄 → Shared reentrancy

Inlining: ➡➡ → Inlined

2.3.3. Module Start/Stop calls

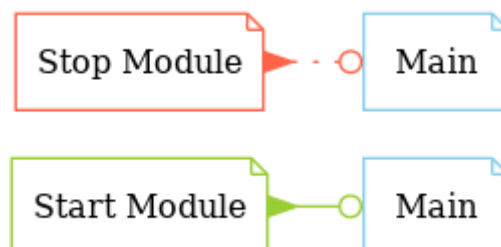


Table 7. Start and Stop module callers

Function	Callers
Start Module	Main.vi
Stop Module	Main.vi

2.3.4. Module Helper Loops

Table 8. Helper Loops Found

Name	Helper Loop Type	Events Handled
No Subdiagram Label	Custom	Timeout User events.Wake HL User events.Sleep HL NI DAQ Request Events.Stop Module

2.3.5. Module custom errors

TIP | Custom errors are added to the module via vi named ***--error.vi**.

Module NI DAQ.lvlib use the following custom errors:

Table 9. Custom errors

Name	Code	Description
Module Running as Singleton	403680	The "%s" module is currently running as singleton, but the Start Module VI was called with 'Run as Singleton' specified as FALSE.
Module Not Stopped	403682	The Stop Module VI for the %s module timed out while waiting for the module main VI to stop. The module main VI may still be running.
Module Not Synced	403683	%s Module was unable to synchronize events.
Module Not Running	403684	Not a single instance of "%s" Module running.
Module Running as Cloneable	403685	The "%s" module is currently running as cloneable, but the Start Module VI was called with 'Run as Singleton' specified as TRUE.
Request and Wait for Reply Timeout	403686	%s
Master Reference Not Closed	403687	The "%s" module cannot be run as singleton because the Master Reference is still open from a prior run as cloneable. If you plan on running this module as both singleton and cloneable, consider changing your Main VI to wire a value of TRUE to the 'Close Master Reference' input of Init Module.vi.

2.3.6. Module Constant VIs

Table 10. Constant VIs Found

VI Name	Data Type	Value
Module Name—constant.vi	["String","String"]	[" %d","NI DAQ"]
Module Timeout—constant.vi	I32	5000
Channel list header ---constant.vi	Array	# Dimentions: (1) - Data Type: (String)

Chapter 3. Libraries

This section describes the libraries contained in the project.

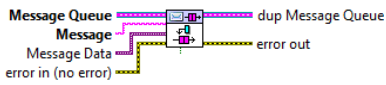
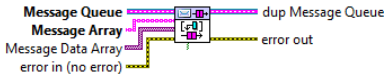
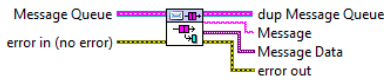
3.1. Message Queue.lvlib

Responsibility: No description found (add content in lvlib description)

Version: 1.0.0.0

3.1.1. Functions

Table 11. Functions (non private scope only)

Name	Connector pane	Description	S.	R.	I.
Enqueue Message (Single)		For a regular message, enqueue the message. For a priority message, enqueue it at the front of the queue.			
Enqueue Message (Array)		For a regular message, enqueue the message. For a priority message, enqueue it at the front of the queue.			
Create All Message Queues		This VI creates and initializes all the message queues for your application. If your application requires more than one message queue, right-click the "Message Queues" constant above and select Open Type Def. Add the new Message Queue(s) to the type definition, then initialize them in this VI.			
Obtain Message Queue		Create a queue with the Message Cluster data type.			
Dequeue Message		This VI pulls messages off the Message Queue.			

Scope:  → Protected |  → Community

Reentrancy:  → Preallocated reentrancy |  → Shared reentrancy

Inlining:  → Inlined

3.1.2. Library Constant VIs

NOTE | No Constant VIs Found

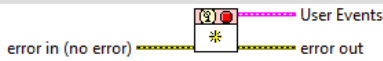
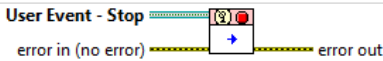
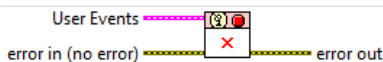
3.2. User Events.lvlib

Responsibility: No description found (add content in lvlib description)

Version: 1.0.0.0

3.2.1. Functions

Table 12. Functions (non private scope only)

Name	Connector pane	Description	S.	R.	I.
Create User Events		Create a boolean user event that will be used to stop the event loop from the message handling loop.			
Fire User Event - Stop		Send the Stop event to stop the Event Handling Loop.			
Destroy User Events		Destroy the user event, even if there is an incoming error.			

Scope:  → Protected |  → Community

Reentrancy:  → Preallocated reentrancy |  → Shared reentrancy

Inlining:  → Inlined

3.2.2. Library Constant VIs

NOTE | No Constant VIs Found

Chapter 4. Custom errors

TIP

Custom errors are added via vi named `*--error.vi`.

Table 13. Custom errors

Name	Code	Description	Owned by
Module Running as Singleton	403680	The "%s" module is currently running as singleton, but the Start Module VI was called with 'Run as Singleton' specified as FALSE.	NI DAQ.lvlib
Module Not Stopped	403682	The Stop Module VI for the %s module timed out while waiting for the module main VI to stop. The module main VI may still be running.	NI DAQ.lvlib
Module Not Synced	403683	%s Module was unable to synchronize events.	NI DAQ.lvlib
Module Not Running	403684	Not a single instance of "%s" Module running.	NI DAQ.lvlib
Module Running as Cloneable	403685	The "%s" module is currently running as cloneable, but the Start Module VI was called with 'Run as Singleton' specified as TRUE.	NI DAQ.lvlib
Request and Wait for Reply Timeout	403686	%s	NI DAQ.lvlib
Master Reference Not Closed	403687	The "%s" module cannot be run as singleton because the Master Reference is still open from a prior run as cloneable. If you plan on running this module as both singleton and cloneable, consider changing your Main VI to wire a value of TRUE to the 'Close Master Reference' input of Init Module.vi.	NI DAQ.lvlib

Chapter 5. Legal Information

5.1. Document creation

This document has been generated using the following tools.

5.1.1. Antidoc

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Maintainer website: [Wovalab](#)

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5.1.2. AsciiDoc for LabVIEW™

Project website: [AsciiDoc toolkit](#)

Maintainer website: [Wovalab](#)

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5.1.3. Graph Builder

Project website: [Graph Builder](#)

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5.2. Product used in the project

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